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Proposal

Requirements

I have been assigned the task of creating and developing a solution for a local green technology company known as **Rolsa Technologies**.

Rolsa Technologies specialises in:

- Solar Panel installation and Maintenance
- Electric Vehicle (EV) charging stations
- Smart home energy management systems

The clients would like a digital solution to be developed that will provide customers with information regarding the following:

- Current green energy products on the market
- How to reduce their carbon footprint

And will allow customers to:

- Schedule consultations and installations
- Calculate their carbon footprint.

Additional market research with existing customers has been conducted, identifying features such as account registration to allow customers to manage their consultations and data, accessibility features to support a wide range of users and a tool for calculating and tracking energy usage. These features can be included within the solution.

In this proposal, I'm going to explain and show how I can create a digital solution that meets the clients needs and any existing or potential users. Below is a full breakdown of how the solution will be created.

Business Context

A few things to consider before creating a proposal, I need to take into consideration what the company provides to see what needs to be implemented as a digital solution. Rolsa Technologies specialises in:

- Solar panel installation and maintenance
- Electric vehicle (ev) charging stations
- Smart home energy management systems

In order to implement an API, I should consider using API Ninja's vast library of free to use APIs. I could use: https://api-ninjas.com/api/evcharger, which finds the nearest EV charger based on location which would be a nice feature to implement into the website.

The reason I chose api-ninjas EV charger API is because it allows clients to find any local EV charging stations near them easily. This would attract more visitors to the website, increasing exposure. Additionally, the API is free up to the first 3000 api calls per month, making it easily scalable in the future through paid plans whereas API alternatives such as carbonfootprint.com requires you to partner with them to even get an API which is a time consuming and potentially expensive process. The same holds true for any other alternatives that would be much better, however, due to a lack of budget / funding, my solution needs to use a public license, free to use assets such as icons, apis etc.

Knowing this, as well as knowing that the company would like to allow customers to schedule consultations and installations, and calculate the footprints, as well as providing information regarding current green energy products on the market and how to reduce carbon footprint, I will have to include a system / way of booking a consultation for either one of the 3 products / services they provide. I'll also have to include a way / page to calculate the carbon footprint of whatever the user inputs.

In order to meet the clients requirement of providing customers with information about current green energy products on the market, I think the best way would be to redirect the users to professional grade, recommended products by the business as a sort of referral as the business is not selling the products themselves. In order to do this, I'll create a little section within the home page that's titled "Recommended Products" with a list of the best green energy products for general use.

Additional things that will be required is an account registration system to allow customers to manage their consultations and data, as well as a variety of accessibility features to support a wide range of users.

In order to calculate the energy consumption, the calculation (wattage * hours) can be used to give a general / rough estimation of how much energy or power an appliance is using. This can be divided by 1000 to convert into kWh. It is then possible to calculate the cost to run the appliance for x hours by taking the kWh usage and multiplying by the cost per kilowatt of energy consumed.

To calculate the carbon footprint, the user needs to identify the source of the carbon footprint first. These are things such as energy use, transportation, food consumption or goods and services.
Secondly, the user will need data such as energy bills, fuel consumptions, flight details, diet information or shopping habits.
Finally, applying the emission factors. Each activity has its own specific emission factor in kg CO_2e per unit. For example, electricity has a factor of 0.4kg per CO_2e per kWh whereas gasoline has a factor of 2.31kg CO_2e per liter or beef has 27kg CO_2e per kg (this varies by country). Your carbon footprint calculation is then calculated by the activity amount multiplied by the emission factor, giving you the equation $CF = A * E$ where CF equals to carbon footprint, A equals to activity amount and E equals to the emission factor.

Empathy Map

Says - I need information regarding solar energy - How could I book a consultation to potentially install energy efficient products at my home? - Can I manage my consultations easily online somewhere? - I want to be able to calculate my carbon footprint somewhere	**Thinks** - I wish I had access to a website that gives me all the information I need before purchasing a product - It would be so much easier to book a consultation online rather than having to wait hours to go to an office or book via telephone. - I wish I could cancel consultations in case I can't make it there. If I had an account maybe I could easily manage it from there? - I wish there was a place somewhere I could easily calculate my carbon footprint from.
Does - Doesn’t end up purchasing a product as its too far away / too much of a hassle to purchase the service - Isn’t able to make it to the consultation because they weren’t able to cancel it, wasting the business’s time.	**Feels** - Annoyed, - Frustrated - Trying to think of an alternative

User Stories

As a [user]...	I want…	So that…	Acceptance Criteria
First time customer	To visit a website that gives me all the information I need about my desired green product (solar, wind, ev etc). Additionally, I want to be able to calculate my carbon footprint	I can compare to competitors to see whether the pricing is over priced or not, and if the quality of the product is good, such as how long it'll last, and its efficiency rating. I need to measure my carbon footprint so that I can see if I'm doing too much of something and if its unhealthy or unsustainable	The website should display any relevant information in easy to spot places so that users don't get fed up having to look for information and end up leaving the website. Should have easy navigation to wherever the user needs to go. A carbon footprint and energy consumption calculator should also be present to allow for easy calculations that the user can do themselves.
Returning Customer	I want to be able to create an account and manage all my consultations and data in case things need updating so I don't have to spend a lot of time sorting out any potential issues.	I can save time on managing all important pieces of information	The system should have a function log in / sign up system that allows users to manage their consultations and accounts. This, in turn, allows for the user to easily be able to manage their account settings, options and data such as consultations, name, email address etc. This is important as it can save the company time in case a cancellation occurs.
Customer with accessibility issues	I want to be able to view the website with ease and not have to worry about any of my potential	I can enjoy the website and gather the information I need the same way as any other user	The system should offer a variety of different accessibility features such as high / low contrast

	accessibility issues making a problem.	would, without having to go through any pain points	options or smaller / larger text. This will allow all users to be able to experience the website in the same way and not have to endure any pain points.

Functional Requirements

The solution should consist of an easy to navigate system with all the important information and features being easily accessible.
It will require the following (subject to change):

- Home page
- Account Registration page
- Account Management page
- Carbon Footprint / Energy Emissions calculator section

The home page will be present upon loading the website. It should present the user with a cookie policy that they can either accept or deny allowing users to have easier access to the website in the future if accepted. It will present the user with a fully working page that consists of a navigation bar, main body, and footer. Within the navigation bar, there will be redirects to all major parts of the website (Home, Account, Calculator, etc) as well as an option to toggle the accessibility menu so that it's easy to see (It will pop out of the left side of the screen once toggled). Under the navigation bar, the main content will provide a redirect to booking a consultation, then a carousel under that that shows all the products that the company offers (solar panels, electric vehicles, smart home energy systems. Only these as they are the only ones mentioned within the brief, however, the code will be made modular and robust so that it is easy to implement new products later on.) This carousel will provide a short description of the product, with the option to find out more by pressing onto the “find out more” option within the product box. For ease of use, I think it's best to place all relevant information on one page, and separate it into sections, (booking a consultation at the bottom, calculator in the middle etc.), having only separate pages for account registration or account information as those are individual to each user, whereas the other content is equally visible to every user. Another feature that could be implemented into the home page is an API that returns the location of nearest public EV charging stations so that users can decide if they need to purchase a home charging station or not.

The inputs for the home page will consist mainly of redirects to a different page, or to a different section, and also one form that can be sent off using POST in PHP for security reasons as no user should have access to it once it's sent off.

The Carbon footprint / Energy usage calculator section should present many options that the user can fill in to calculate their carbon footprint or energy consumption. It will consist of a form and selection box, the user selects the type of calculation they wish to have, and then then fill in the corresponding values, with the maths being processed behind the scenes and displayed to the user. Within here, there will be a button that users can press which asks a generic question such as “How can I reduce my carbon footprint?”, when prompted, a little text box will show up on the users screen providing valuable information on how the user can decrease their carbon footprint with ways such as:

- Using sustainable travel through the forms of walking, biking, public transportation
- Making dietary changes (eating less meat, more plants, locally, making less waste)
- Reducing consumption (reuse, recycle, buy less, choose sustainable products)

The account registration page will be a redirect from the main home page, where the user will have to fill in information such as Name, Email address and a password to create an account. This password should have validation such as over 8 characters in length, and including at least a number and special character as to comply with GDPR, which will then be securely hashed using an algorithm such as SHA-256 as it has strong data integrity and security. Once the form is completed, it is sent to the backend database where it is securely kept for future reference and use.

The account management page can be accessed under the account tab which will be under the navigation bar. Before a user can access it, they will have to log into their account once they've created it using the account registration page, and the same credentials. The passwords are compared to the same field using the email address as a key, and if the hashed passwords match, that means that it's a secure login / valid login and the user can proceed to access all their information. Here, all the users bookings will be available, with the user being able to update information such as cancelling bookings, or updating their passwords / names and important information. Currently, I won't be able to implement an email verification / 2fa system as that would require an email server, which I do not have access to. As a result, a future consideration would be to include a 2 factor authentication system when trying to log in to further increase the system's security.

By doing this, I can ensure that any potential risks are mitigated and the solution would be in accordance with the law.

Non functional Requirements

Security

When creating the system, I need to ensure that the system is secure and robust. In order to ensure this, a checklist I must follow is:

- System must validate all data inputs
- System must use secure cookies
- System must use sha256 encryption for passwords

This is so that the system is in compliance with laws and regulations such as GDPR, using sha256 for safe and secure encryption. By implementing both syntactic and semantic input validation, it can prevent injection attacks. Additionally, I can use PHP PDO to automatically handle most inputs securely, however, the more checks that are present, the better as they have minimal performance effect, but greatly increase security.

Additionally, like I mentioned before, it's important to ensure that passwords are strong as hashed passwords alone are not 100% secure. It is ideal to combine with a strong password such as one with 8 characters, at least 1 number and special character. This will provide the system with a strong security combination.

One improvement I could consider is to add a password deny-list to avoid commonly used passwords or ones that have been involved in a data leak. This will further reduce the likelihood of a system being vulnerable to attacks. Additionally, I could also limit the number of sign-ins to a maximum of 3 per user per x time slot. This would prevent brute forcing.

Following secure cookie management as per the UK PECR and UK GDPR, it is important to obtain explicit consent for cookies that are non-essential, giving the user the option to easily withdraw. These cookies should also be stored securely through things like httponly, or same site flags, ensuring cookie policy transparency and maintaining clear audit logs.

Maintainability

It is important that after the solution is developed, that it is easy to maintain and upkeep.
In order to ensure good maintainability, only proper coding practices will be followed throughout the code. This means that my code will have a clean and modular code structure, where each component (e.g., navigation bar, calculator, account management) is independent and reusable, avoiding redundant code and it will have consistent coding standards such as PSR for PHP, or W3C for HTML / CSS, where meaningful variable and function names are present, giving proper identification to what they are doing / their intended use, having consistent and proper indentation and clear commenting.

To make it even more maintainable, version control should be used throughout the process so that in case things go wrong, the solution can be reverted to a prior, working version. These commits should be regular (ideally once something new has been implemented) so that no large gaps are present in between commits.

If properly followed, these few simple things will allow future development and maintenance of the solution to be very simple

Performance

Any good solution should ensure that the system is fast, responsive and resilient under typical and peak usage so that the systems run without any issue.

Using User-Centric metrics, a website should load under 200ms and any Authentication / Account endpoints shouldn't take longer than 300ms with external API calls being between 300 - 800ms and timeout over 2s.

The system should demonstrate compliance with all targets.

Usability -> Accessibility

The system should be straightforward to use for both the end users and the content editors that may oversee any of the backend systems.

In order to achieve this, the system should be adapted to different user needs, following the WCAG guidelines.

The guidelines are as follows:

- Perceivable: provide text alternatives for all non-text content, ensuring that captions and transcripts are available for all multimedia and that the color contrast has a minimum of 4.5:1 for text.
- Operable: all functionality must be accessible via keyboard navigation, providing visible focus indicators for interactive elements and avoiding time limits or providing an option to users to extend them.
- Understandable: Using clear and simple language with predictable UI behaviour to prevent confusion, providing error identification and suggestions clearly and maintaining consistency throughout the layout across all pages.
- Robust: All code must be validated against 23C standards and ensuring compatibility with assistive technologies such as screen readers and voice input

Decomposition

The main page will consist of a few inputs. Most notability, the navigation bar and two forms present within the page.

In order to have a fully working solution that follows the brief, I've opted to make the main page contain all the content. This makes it easy to create and optimise as I can keep all relevant files together such as the CSS, making it modular, robust and easy to understand.

The main page will be split up into 6 sections, the home section, about us section, products section, consultations section, calculator section and footer. These can be accessed either by manually scrolling through the page, or by using the top navigation bar and pressing the corresponding section, Although, the home page and about us section will be merged into one as they are more or less the same thing, just an addition of one another.

In order to book a consultation or installation, the user will have to fill out a form under the corresponding section (consultation section) with all correct information. These will be validated through methods such as regex to ensure valid telephone numbers or UK postcodes as the company is assumed to be working from within the UK (local). This will ensure that many forms get denied if they do not have correct information, ensuring verifiable data can be stored. Other forms of checks would be presence checks for special characters such as an "@" followed by correct email types (e.g., gmail.com or yahoo.com) or length checks to ensure that values do not go above a given length (e.g., 20 characters for a first name). The user can then check their history from within their account as the booking will be linked via their email address.

For the calculator, users have to fill in the form with corresponding values. This will be easy to validate as most of the form will be standard drop down values, not allowing any user input, however, in some cases, the user may have to input numbers. This can be checked via a value check to ensure that the entered string is only an integer or float (depending on the situation) or where the second value cannot be greater than the first value etc.

Once the user tries to access the "Your Account" page, if the user has not previously logged in, they will be prompted to log in, or create an account which can be redirected to from the log in page and vice versa. These login procedures follow the standard validation, using regex similar to the consultation booking for emails or length checks for names. The user will be asked to enter their password twice, using encryption (sha256) when storing the information. If the user has already logged in, the next time they enter the page, it will automatically allow the user to view and manage their account data and history. All prior bookings will be shown here with a status, and if the status is not completed or cancelled, it will be pending, allowing the user the action of cancelling the appointment. The user will also be able to edit their name and any other information as per GDPR regulations, however, I will not be able to set up email verification or changing as it could prove to show more security concerns and errors as I cannot set up an email server due to a lack of infrastructure.

Finally, for the accessibility options, the user will be able to access a menu in the bottom left corner of the screen that is constantly present. This menu will provide the user with options such as larger / smaller text, or high / low contrast which will make it more accessible for users.

KPIs

Business KPIs

Business Key performance indicators are important for a company to have as they can help indicate what is going well and what needs to be improved. As a result, a company can

potentially improve many aspects of their business based solely on statistics.

Some important KPIs are:

- Reviews: Reviews from sites such as trustpilot can help attract new customers and can tell the company what they are doing well / what users generally think of a website.

- Visitor Satisfaction Rate: These can help in identifying areas of improvement and therefore boost customer's overall experience with the website. Industry standard is set to a 80% satisfaction rate and higher, however, the higher the overall rate is, the better the company is doing as it means the users are having an overall better user experience. These can be tracked through short surveys of the users overall experience on the website, however, many users tend to just skip over the survey completely / are not bothered to properly read what the questions are asking. This could lead to invalid data.

- How many users return to the website: This can be measured through sessions and calculating the time between visits. This is useful because a business wants to be as repetitive as possible to be able to reuse customers. If customers are more likely to come back, that means they really enjoy the service, and are more likely to refer others / purchase from the company again.

Website KPIs

Website key performance indicators track how performative a website is, allowing the company to make changes based on data to see what they can improve on in order to increase users and user engagement.

Some important KPIs are:

- Page load speeds: As mentioned before, it's important to measure how fast a page loads so that users do not get fed up whilst waiting for a page to load. Studies have shown that people perceive time subjectively, and waiting for a slow website feels longer than the actual time passing, being 15% slower than they actually are and remembers them as being "35% slower" later. It also forms a strong negative first impression between the user and company if a website takes too long. If a page loads within milliseconds, users tend to associate it with a sign of credibility and professionalism, whilst slower sites imply the opposite. Lastly, in a market where you have to compete against sites with high-performance such as google and amazon, you expect all modern sites to load instantaneously.

- Bounce Rate: This shows the company the percentage of users that visit the site just to leave immediately without taking any actions. Logically, if the bounce rate is high, then the site is performing badly and not capturing the attention of the end-user, therefore the site may need updating. This is a crucial tool for a company to have because the company can make decisions such as, for example, if users visit the site and leave immediately, it shows that the main home page may not appeal to what the users had in mind. This means that the home page might not have the correct

information and could be losing the company customers. As a result, the home page can be updated to see if the bounce rate goes down, and if it does, then the new website attracts more visitors than the old one, potentially increasing the company's clients.

- Conversion Rate: This indicates how many of the users that actually visit the site purchase a product / reserve a consultation. It is important that this number is reasonably high so that the business is making money. A higher conversion rate means that the company is able to transform more clients into customers whereas a low conversion rate may tell the company that their products or services are doing well. This could lead to the company deciding to remove a service or product altogether as you can monitor conversion rates for individual products, showing the company what is performing the best.

Final Justification

The proposal shown above was all my decomposition, analytics and requirements for the system to run as the client wants. In this final justification, I will explain why I've decided to do everything this way.

My proposed solution is an inclusive website that's able to suit all users, ranging from those with accessibility issues to those without. I've ensured that the solution will meet all accessibility guidelines following W3C and WCAG, decomposing how I'll be following the guidelines when creating the proposed solution. This meets the needs of the clients and users by providing a wide variety of accessibility features to support a wide range of users which was part of the market research conducted by the client gathered from existing customers.

In addition, the proposed account management and registration system will be easy to use and understand, providing error messages if something goes wrong and allowing users to easily manage their account details and stored data in a way that does not break anything. This meets the requirements of both the clients and users as it provides a feature that allows account registration to allow customers to manage their consultations and data.

Finally, the proposed design including information and tools for calculating energy usage / carbon footprint meets the remaining requirements as it provides a tool for calculating and tracking energy usage, a calculation for the users carbon footprint, scheduling for consultations and installations and giving important information on how to reduce the users carbon footprint.

How potential risk might be mitigated

Any risks found such as security concerns have been addressed as they will follow secure GDPR guidelines to ensure that all potential problems have been mitigated.

In addition, I need to take under consideration potential issues with informational inaccuracies. In the slight chance that any potential information gets edited and gets corrupted or damaged, this could lead to the company losing valuable information regarding customers or potential business information. To mitigate this risk, it's best to hire a dedicated staff member to oversee all the data flows within the system, and potentially think about implementing a CMS (Content Management System) ; however, for the scope of this project, it's outside of my budget and time constraints to implement something of that scale.

The same could be said for any potential data loss that may occur during the consultation booking process. Many things could occur here, such as the process being overcomplicated or not having enough customer support, but the main issue is any potential data loss / corruption as the business may lose its schedule for when it's supposed to consult clients. In order to mitigate this risk, the system must be thoroughly tested to ensure that no sql injections can occur and that only valid / validated data can be passed through. This however, cannot fully prevent data loss by itself. Therefore, it's important to keep backups of data just in case something does happen to go wrong.

Conclusion

In conclusion, this solution meets the different user and client requirements as well as the task brief requirements, both functional and non functional, whilst simultaneously taking into account any potential risks and how to mitigate them.

Research

Hardware within the Industry

Some common industries that a green technology company may be included within are renewable energy, sustainable agriculture, waste management & recycling, smart city solutions and electric vehicles. All of these industries focus on environmental solutions like reducing emissions from vehicles or electric waste, conserving resources or improving efficiency.

Solar Panels

Hardware such as solar panels are used throughout the industry in order to decrease emissions from decreasing reliance on fossil fuels as well as providing clean, renewable energy that significantly saves customers on electricity bills and decreases carbon footprints.

Solar panels work by generating electricity through photovoltaic cells, usually made out of silicon, that absorb sunlight (photons). Due to absorption of the photons, the electrons from within the semiconductor material (in this case, silicon) cause them to break free from their atoms. This in turn, creates a direct current as the free electrons flow in one direction. Once the DC current is generated, an Inverter converts the direct current into alternating current, which is what is typically used within a home.

Due to this, solar provides long-term energy independence, low maintenance and long-term savings as excess energy can be sold off to the grid, or stored for future use which makes them a viable option for customers to use at home as a long-term investment.

Wind Turbines

The same holds true for hardware such as wind turbines. These are widely used across the energy industry to reduce emissions in the same way as solar panels, reducing reliance on fossil fuels, whilst providing clean, sustainable and renewable energy.

Wind turbines work by converting kinetic energy created by wind, the turbine blades rotating in the wind, driving a shaft that is connected to a generator. From inside the generator, this mechanical energy is transformed into electrical energy through the process of electromagnetic induction.

This offers long-term sustainability, low operating costs and energy independence as excess electricity that is generated can be fed back into the grid or stored for later use. However, large wind turbines are typically used for industrial purposes and often are expensive to set up.

Geothermal

Geothermal energy is used in similar applications as wind turbines and solar panels, however, it also allows for the heating and cooling of systems and direct heat applications as it produces a mass amount of heat.
Geothermal works by harnessing the earth's internal heat, using drilled wells to access underground reservoirs of hot water or steam which then drives turbines connected to generators to produce electricity in a similar way that wind turbines use the kinetic energy of wind to spin a shaft, converting mechanical energy to electrical energy.

Geothermal energy is a viable option for sustainable energy and commonly used within the industry because it is clean, renewable and a reliable power source with very low greenhouse gas emissions and small land footprint.

Sustainable IT

Sustainable IT is a generic term for hardware that is less defined by unique components and more by its design, lifecycle and operational characteristics. Sustainable IT is specifically created to have Longevity and durability in mind, using robust materials and design principles to extend its working / useful life, reducing the need for replacements. They are also made to be specifically modular and repairable so that instead of recycling an entire device, certain components can be reused or replaced, reducing waste and extending the total devices lifespan. Additionally, the devices are specifically designed to be energy efficient, consuming menial power during both active and idle periods to increase efficiency and lifespan. All these factors, and a few more, make sustainable IT used widely throughout the industry as it creates longer lasting products that are better for the environment, easily recyclable, easy to upgrade, and less expensive.

Using sustainable IT, you can create products such as efficient processors that are chips specifically designed for lower power consumption, or smart building tech such as electrochromic smart glass, smart appliances, automated sensors and efficient cooling systems for large data centers or other hardware.

Software within the Industry

Some common software used within the green industry are systems such as Sustainability and ESG reporting software, Life Cycle Assessment (LCA) Software and Environmental Monitoring software.

Sustainability and ESG reporting software

These kinds of software / platforms help companies measure, manage and report on their environmental, social and governance performance, including their carbon footprint across different emission scopes. They streamline data collection and ensure compliance with regulatory frameworks such as the global reporting initiative and the EU's Corporate Sustainability Reporting Directive.

They are used to enhance reputation as companies that disclose their ESG metrics often see an enhanced reputation among customers, investors and the wider public, cost savings as ESG initiatives often lead to more efficient use of resources which can result in significant cost savings. For example, reducing energy consumption not only lowers environmental impacts but also decreases utility costs, and a competitive advantage as consumers become more aware of and concerned about sustainability issues.

Some examples of Sustainability and ESG reporting software include Prophix One, Workiva and greenly.

Life Cycle Assessment software

This software is commonly used to evaluate the total environmental impact of a product or service throughout its entire life cycle. Key features include data modeling to track energy usage, emissions, water consumptions and waste at each stage of the product life cycle and scenario analysis to compare different designs or material choices in order to identify the most sustainable option.

These software are used within manufacturing to redesign products for a lower carbon footprint and reduce waste. They are also used in customer goods as brands evaluate packaging options (plastic vs biodegradable alternatives) to minimise costs and environmental impact and construction to help assess building materials and energy efficiency strategies to achieve green certifications.

Examples include One Click LCA, Sustainly and Ecochain

Environmental Monitoring Software

These tools track specific environmental parameters such as air quality, water consumption, noise levels and waste production in real-time. Key features include sensor integration to connect IoT devices for continuous monitoring, alerts and dashboard to send notifications when thresholds are exceeded, and regulatory compliance to ensure adherence to local and international environmental standards.

These software are commonly used within manufacturing plants to monitor emissions and compliance with environmental regulations to avoid fines such as tracking volatile organic compounds in chemical plants as well as energy and utilities to track air quality and water discharge to be within legal compliance. Additionally, they can be found in smart city infrastructure as the tools monitor pollution and noise levels, water quality and others to help improve urban sustainability and public health.

Examples include SafetyCulture, Envirosuite, and ERA EH&S

Newly emerging technologies within the industry

Hydrogen and Electric mobility

These typically refer to the usage of hydrogen or batteries to power vehicles, offering clean alternatives to fossil fuels with Battery Electric vehicles using stored electricity and Hydrogen Fuel Cell Electric Vehicles generate electricity onboard from hydrogen. These are newly emerging technologies with electric vehicles already being a popular option today, whereas the hydrogen powered vehicles lack the infrastructure to be made mainstream.

Advanced Recycling

This is the process of breaking down used plastics into their basic chemical building blocks such as monomers, oils and gases using heat or chemicals, allowing them to be reformed into new, high-quality plastics.

This is a beneficial new technology as it introduces a new way of breaking down traditionally “hard-to-recycle” plastics, allowing for higher recycling rates and less waste as everything can now be reused.

AI & IoT

This is both AI and IoT. IoT is a network of physical objects such as devices, sensors and vehicles with embedded software to collect and exchange data over the internet. These are used as they gather vast amounts of real-time data from the physical world and can be used to measure pollution levels and similar.

AI is a computer system designed to perform tasks requiring human intelligence, making predictions by finding patterns and deciding on actions.

They work together by collecting data and forming decisions based on that data. Commonly used within smart homes where the thermostat learns your preference and adjusts automatically or healthcare where wearable monitor vitals alert doctors of irregularities.

Legislation and industry specific guidelines

UK GDPR & Data Protection Act 2018

Governs how you collect, store and use personal data. This requires:

- Transparency (telling the user what data is being collected and why)
- Lawful basis for processing (consent, having a valid reason for processing data)
- Security measures to protect data, such as implementing technical or organisational safeguards to protect data from breaches or misuse like hashing
- Giving users rights like access, correction and deletion to their personal data.

Privacy and Electronic Communications Regulations (PECR)

This covers electronic marketing and cookies. In order to comply with the regulations, I must get consent before using non-essential cookies (analytics, ads) and provide clear cookie notices and options.

E-Commerce Regulations

Because the company is selling a product / service online, it must display the company name, address, registration number, VAT number and contact details.

Equality Act 2010 & Accessibility

THe website must be usable by people with disabilities. This means that the website should follow WCAG standards (e.g., text alternatives for images, keyboard navigation) and must publish an accessibility statement.

CMA Green Claims Code

Since the company is a “Green” company, if it makes claims such as “eco-friendly” or “sustainable”, they must be truthful and clear, supported by evidence and cover the full lifecycle of the product, not just one part.

ASA/CAP Advertising Guidance

Advertising rules for environmental claims must avoid vague or absolute terms unless fully proven, must not be misleading by omitting important details, and provide context for claims like “carbon-neutral”.

Environmental Regulations & Net-Zero goals

The products and operations should align with the climate change act of 2008 and environmental act, as well as UK's net-zero targets. If the company deals with renewable energy or low-carbon tech (which it does), extra safety and emissions standards may apply.

Reference website:
https://www.youtube.com/watch?v=OiJaX3VJox0